

yadiggg: Adversarial Testing & System Validation Protocol

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1. Introduction: The Adversarial Philosophy

Engineering a premium, offline-first product like **yadiggg** requires challenging every architectural and hardware assumption before finalizing the design. If a system is not tested under extreme, adversarial conditions, it *will* fail in the field—whether due to physical wear, low-light environmental scanning, database index fragmentation, or inter-process core synchronization bugs.

This document serves as the official **Red Team Verification Protocol** for **yadiggg**, detailing the stress-tests and validation steps designed to break our assumptions and guarantee a bulletproof commercial rollout.

2. Software & Database Stress-Testing (The "Red Graph" Tests)

2.1 Assumption 1: Local SQLite query latency is sub-50ms for a 1,000,000-release database.

- **The Adversarial Reality:** In the real world, database indexing suffers from fragmentation over time, and highly connected networks (such as tracking sample relationships or deep session musician credits) can result in slow recursive joins. SQLite has limited query-plan optimization compared to server-side relational databases.
- **Adversarial Step (Test DB-01 - "The Credit Avalanche"):**
 1. Generate a dummy SQLite database containing exactly **1,500,000 releases**, **12,000,000 tracks**, and **30,000,000 collaborative credit edges** (representing high-density session databases).
 2. Inject recursive sampling lineages where a single loop is sampled across **15 generations** (forcing SQLite to run recursive Common Table Expressions (CTEs) deep down the tree).
 3. Execute simultaneous wildcard searches (e.g., fuzzy name matching like `LIKE '%william%'`) on an ARM Cortex-A53 test board.
 4. **Failure Trigger:** Any query taking longer than **100ms** or triggering more than **10MB** of temporary RAM allocations during index traversals.

2.2 Assumption 2: OCR scanning using Tesseract-lite takes sub-500ms on Cortex-A53.

- **The Adversarial Reality:** Real crate digging happens in dark record store basements, under dim yellow lamps, scanning wrinkled, plastic-wrapped, or faded vintage spines printed in cursive, gothic, or highly stylized 1970s psychedelic fonts. Angled/skewed captures can distort characters, driving character-error-rates (CER) to 100%, causing Tesseract to burn CPU cycles trying to decode visual noise.
- **Adversarial Step (Test OCR-02 - "The Vinyl Basement Fuzz"):**
 1. Assemble a test-suite of **500 challenging spine scans**: blurred captures, low-contrast text on reflective plastic sleeves, extreme lighting gradients, and curved or distorted text.
 2. Write a fuzzing script to artificially add Gaussian noise, random motion blur, and a 25° rotational skew to the test images.
 3. Feed this skewed, degraded data into **Tesseract-lite** running on a single ARM Cortex-A53 core running at a throttled **800MHz** (simulating thermal throttling under hot summer warehouse conditions).
 4. **Failure Trigger:** Character localization taking over **1.2 seconds**, or OCR classification causing a core-level thermal shutdown.

3. Hardware & Firmware Security Penetration (The "Lockbox" Tests)

3.1 Assumption 3: Encrypting the user data partition with AES-256-XTS and storing keys in the i.MX 8M Nano CAAM prevents unauthorized database

extraction.

- **The Adversarial Reality:** A malicious competitor or pirate could buy a physical **yadiggg** device, de-solder the eMMC flash memory, and read the raw binary contents using an external card reader. Alternatively, they could attach logic analyzers to the PCB's JTAG pins or the eMMC bus to intercept key negotiations during boot.
- **Adversarial Step (Test HW-03 - "The Atlanta Heist"):**
 1. Use a hot-air rework station to de-solder the eMMC flash chip from an engineering prototype.
 2. Mount the chip to an external SD-card reader breakout and attempt to mount the data partitions on a Linux development machine. Verify that all relational database tables are completely unreadable without the CAAM-specific hardware key.
 3. Attempt to connect a Segger JTAG debugger to the CPU's exposed JTAG debug pads during bootup to extract the active keystore from LPDDR4 memory.
 4. **Mitigation / Action:** Verify that the NXP eFuse register `SECURE_BOOT` is blown to permanently disable JTAG debugging and close physical memory dumping backdoors in production.

4. Multi-Core & Low-Power Feasibility (The "Crossover" Tests)

4.1 Assumption 4: The real-time Cortex-M7 core can seamlessly stream captured PDM audio to the Cortex-A53 Linux layer without audio pops or memory buffer overflows.

- **The Adversarial Reality:** In heterogeneous multicore systems, inter-process communication (IPC) and memory cache synchronization are highly complex. If the A53 core is waking up from Deep Sleep while the M7 core is actively writing PDM microphone data to shared SRAM over DMA, a cache-coherency conflict can occur, causing audio clicks, memory alignment corruptions, or complete kernel freezes.
- **Adversarial Step (Test FW-04 - "The Glitch Mob"):**
 1. Trigger a constant stream of manual wake/sleep transitions (100 cycles per minute) by spamming the GPIO scan button interrupt.
 2. During transitions, force the Cortex-M7 to stream a continuous, high-volume 1kHz test sine wave from the Knowles PDM microphones into the shared RAM buffer.
 3. Check the captured audio stream on the A53 side for any lost packets, misaligned bytes, or audible popping.
 4. **Failure Trigger:** A single dropped audio sample (which would corrupt the OLAF Sonic ID fingerprint match) or a kernel panic during A53 wake-up.

5. Industrial Design & Thermal Stress (The "August in Atlanta" Tests)

5.1 Assumption 5: The translucent polycarbonate housing can maintain thermal equilibrium without active fans.

- **The Adversarial Reality:** Passive cooling inside a sealed 15mm-thick plastic pocket casing can act as a thermal insulator. Running a quad-core processor at 1.4GHz doing heavy mathematical vector math inside a hot, un-airconditioned record warehouse in the middle of summer can quickly drive SoC temperatures above 85°C, triggering severe thermal throttling or device shutdown.
- **Adversarial Step (Test MECH-05 - "The Warehouse Melt"):**
 1. Place the fully assembled **yadiggg** device inside a thermal chamber heated to 45°C (113°F) with 80% relative humidity.
 2. Trigger continuous scanning loops (running OCR and OLAF calculations consecutively) for **4 hours straight**.
 3. Monitor the internal SoC core temperature sensors and verify the casing's external surface temperature.
 4. **Failure Trigger:** External surface temperature exceeding 42°C (107°F) (making it uncomfortable to hold in the hand) or internal thermal throttling reducing A53 processing clock speeds below **1.0GHz**.

6. Sourcing & Assembly Vulnerabilities (The "Atlanta Sourcing" Tests)

6.1 Assumption 6: Local assembly in Atlanta, Georgia, is financially and operationally feasible using standard off-the-shelf components.

- **The Adversarial Reality:** Relying on single-source local distributors or highly specialized, boutique components can leave the project completely paralyzed by sudden component shortages, lead-time changes, or local manufacturing assembly errors.
- **Adversarial Step (Test OPER-06 - "The Broken Link"):**
 1. Perform a comprehensive Bill of Materials (BOM) risk analysis. For every single integrated circuit (IC) spec' d in the schematics, identify at least **two pin-compatible second-source alternatives** (e.g., swapping the Micron eMMC with a Samsung equivalent, or the TI haptic chip with an Analog Devices counterpart).
 2. Challenge local assembly pricing by auditing Atlanta-based assembly line overhead costs against rapid prototype assembly shops (like Advanced Circuits and Tempo Automation), factoring in a minimum **35% gross margin** at a \$249 retail price target.

7. QA Integration and To-Do Updates

These adversarial validation tests have been formally integrated into our roadmap as **Task 10 (Adversarial Quality Engineering & Validation)**. Setting up this rigorous testing harness ensures we do not build a single line of software or solder a single circuit pad without knowing exactly how to break it!